Office hours
REVISED

- Fridays 12-1pm (zoom or Stewart Biology, N5/8)

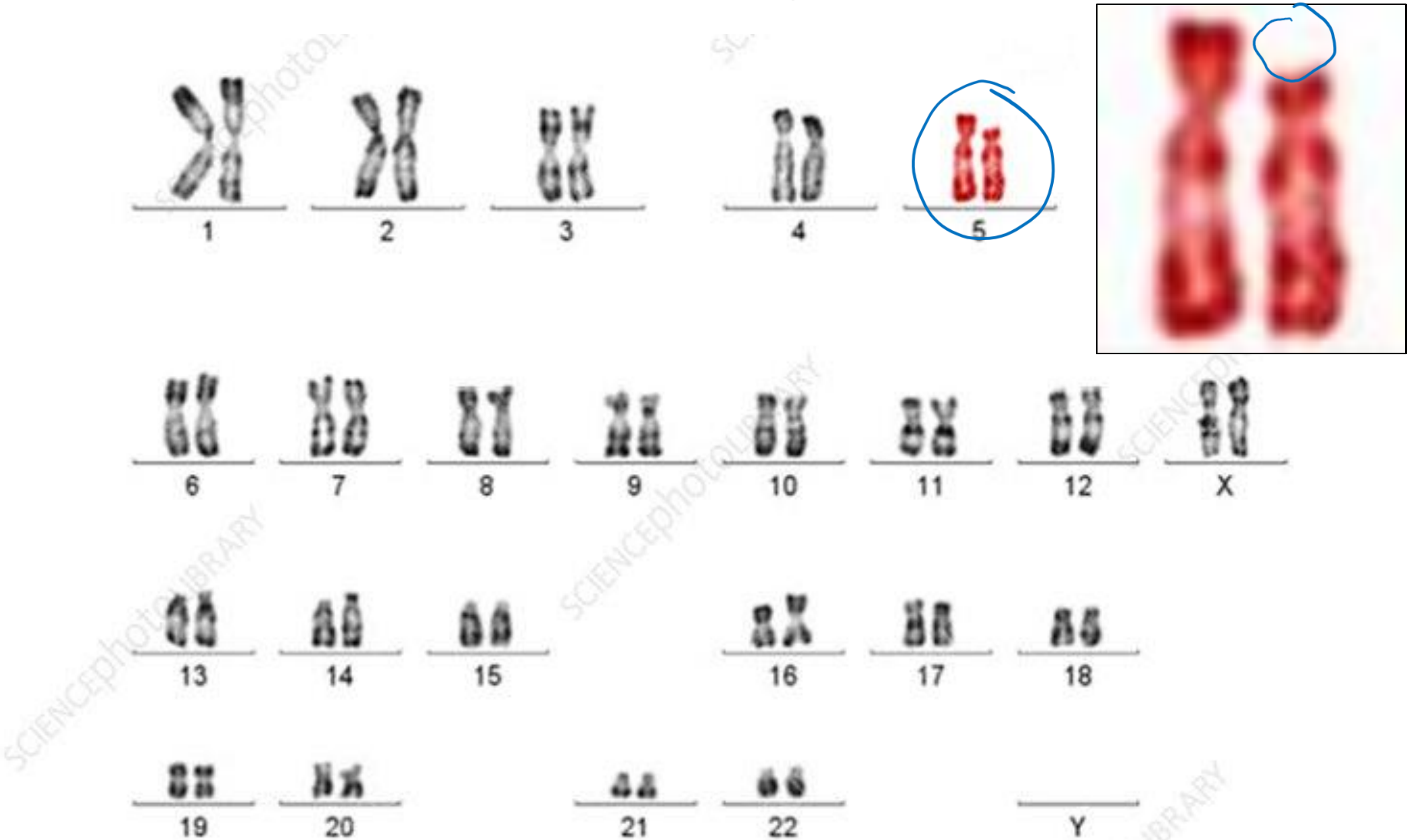
https://mcgill.zoom.us/j/81407275167?pwd=8Cgm0RPnWQRlOdaCNFyXnS3M8amQUj.1

- By appointment
- Before or after class, right here in Leacock 132

Large Scale Chromosomal Changes

Changes in Chromosome Structure

Cri-du-chat Syndrome



Cri-du-chat Syndrome

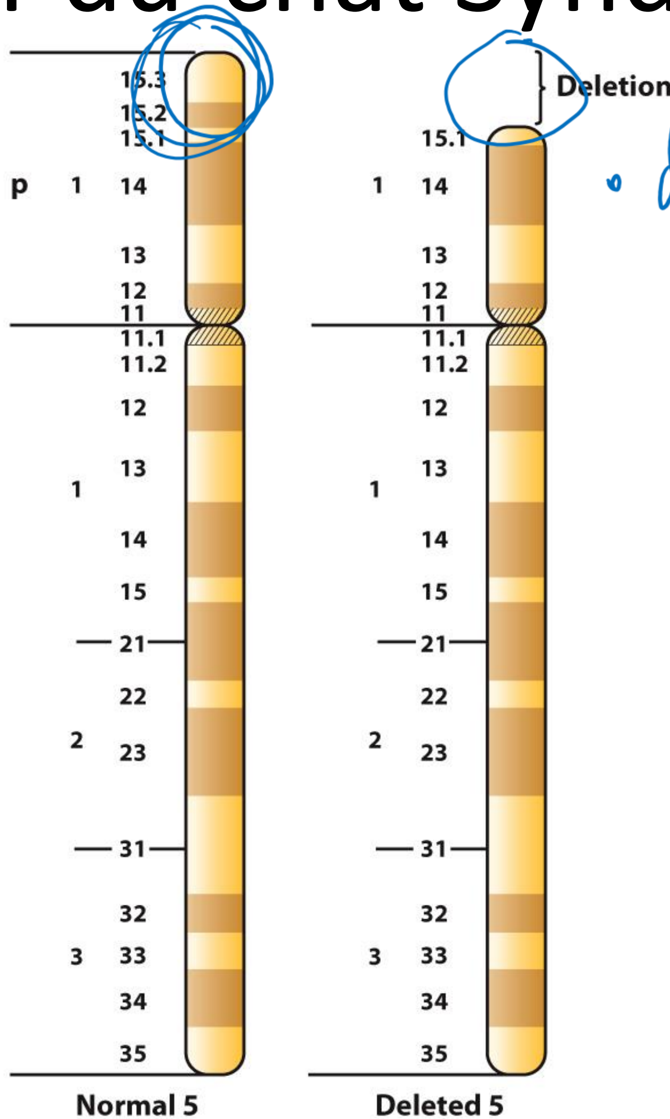
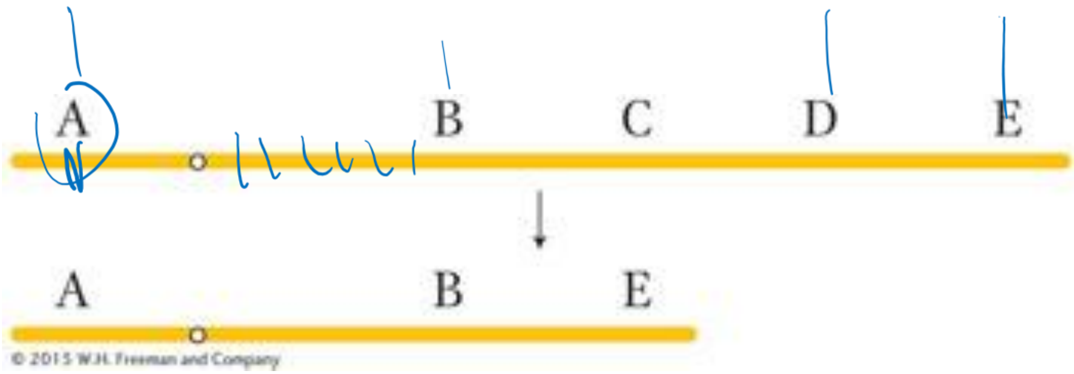


Figure 17-22
Introduction to Genetic Analysis, Eleventh Edition

- deletion contains ≥1 genes important for development
- at least one of these genes is haploinsufficient

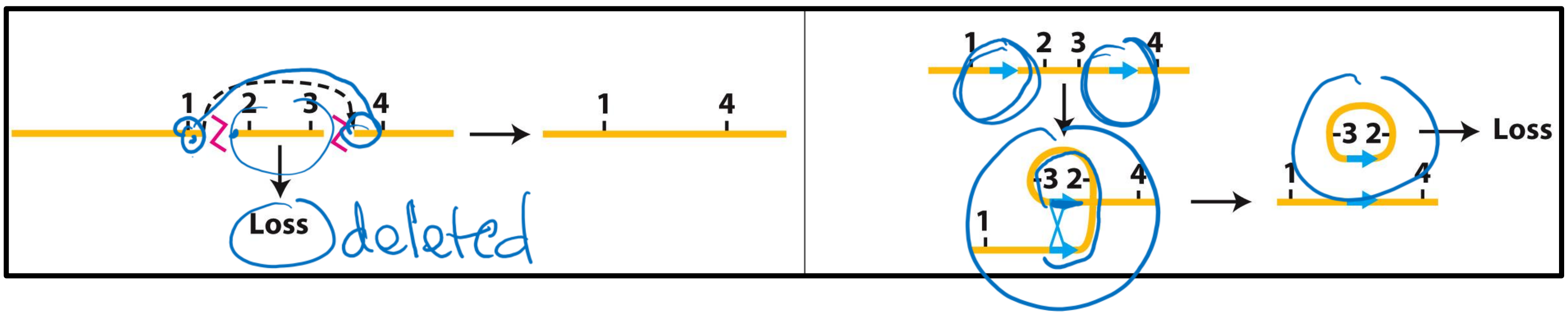
Chromosomal deletions and their possible causes



Can be large
Can be small

How would genes look on this diagram?

Breakage and rejoining | **Crossing over between repetitive DNA**



Chromosomal deletions

example: possible origin of the Williams syndrome deletion

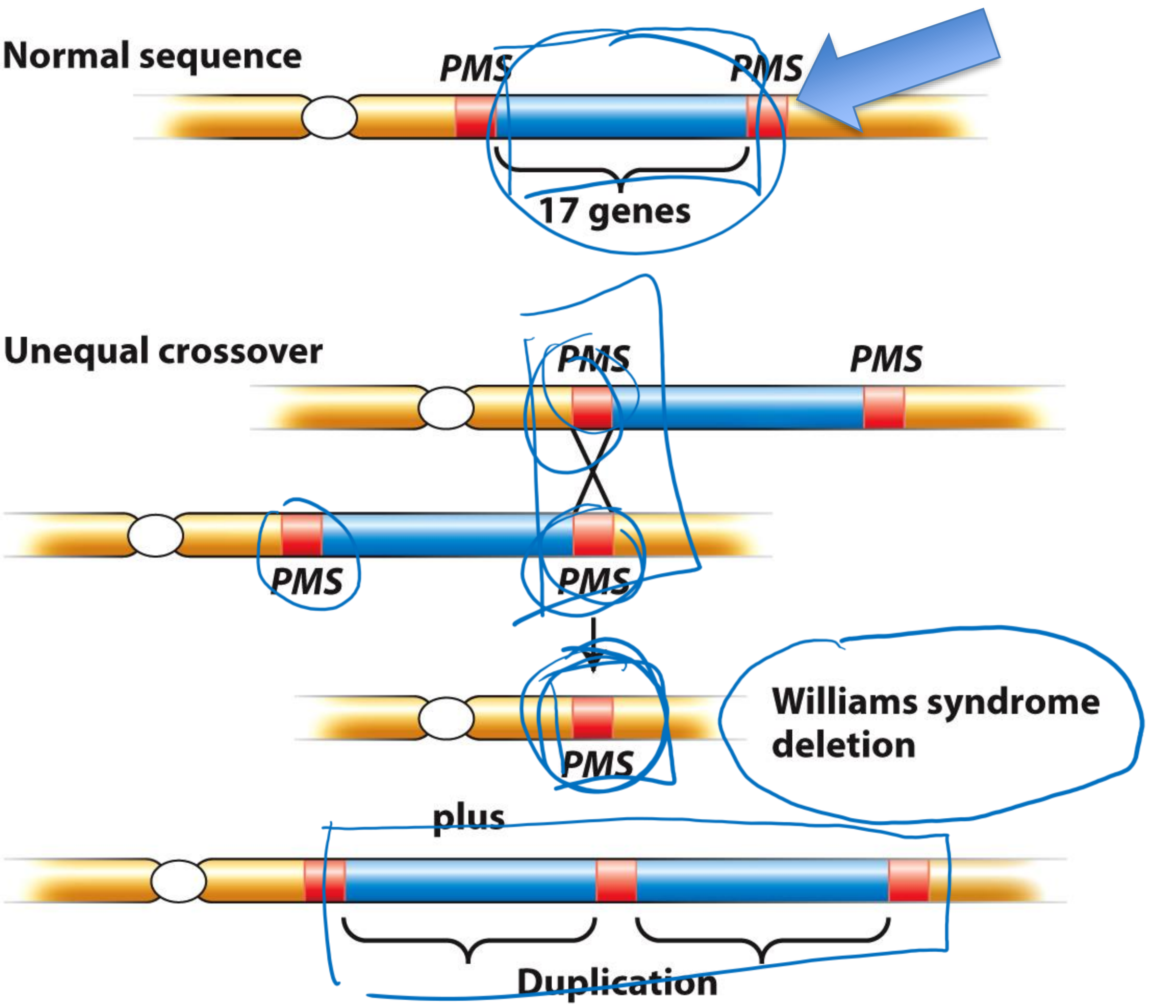


Figure 17-23
Introduction to Genetic Analysis, Eleventh Edition

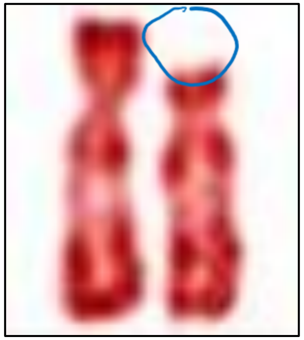
Williams syndrome is found at a frequency of about 1 in 10,000 people.

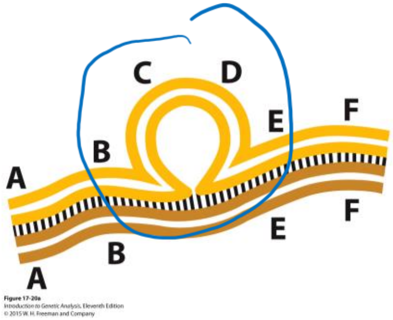
Among other traits, individuals often have pronounced musical or singing ability, as well as hyper-sociality.

The syndrome is almost always caused by a 1.5-Mb deletion on one homolog of chromosome 7, specifically at band 7q11.23.

How to detect chromosomal deletions

By observing their length or pairing



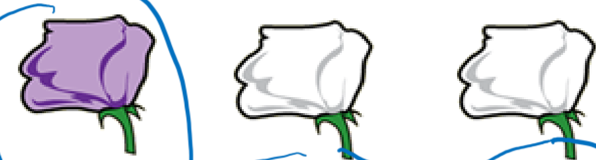


Deletion loop observed in paired homologous chromosomes

By doing a complementation test

Through DNA analysis using genomic techniques

Complementation test - concept



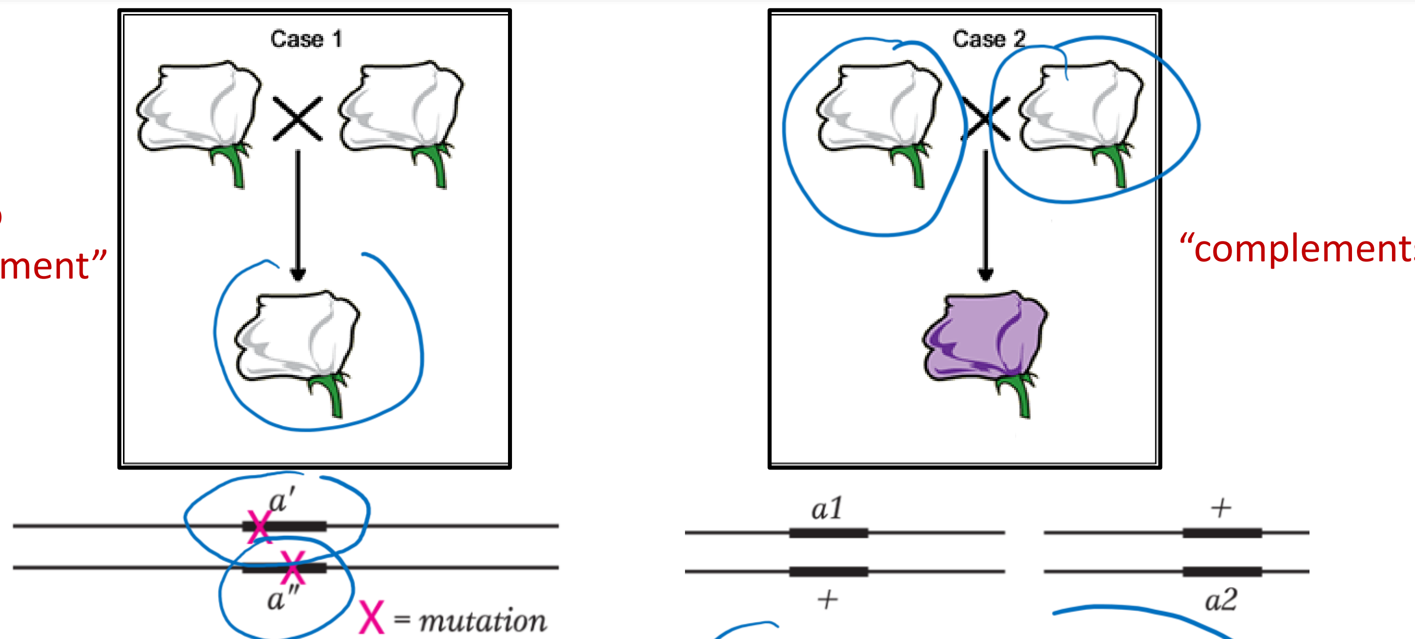
Wild type

Mutant strain 1

Mutant strain 2

Do strain 1 and strain 2 have mutations in the same gene?

"fails to complement"



"complements"

Interpretation: Strain 1 and strain 2 have mutations in the same gene

Interpretation: Strain 1 and strain 2 have mutations in different genes

Testing for chromosomal deletions using complementation

Strain to be tested

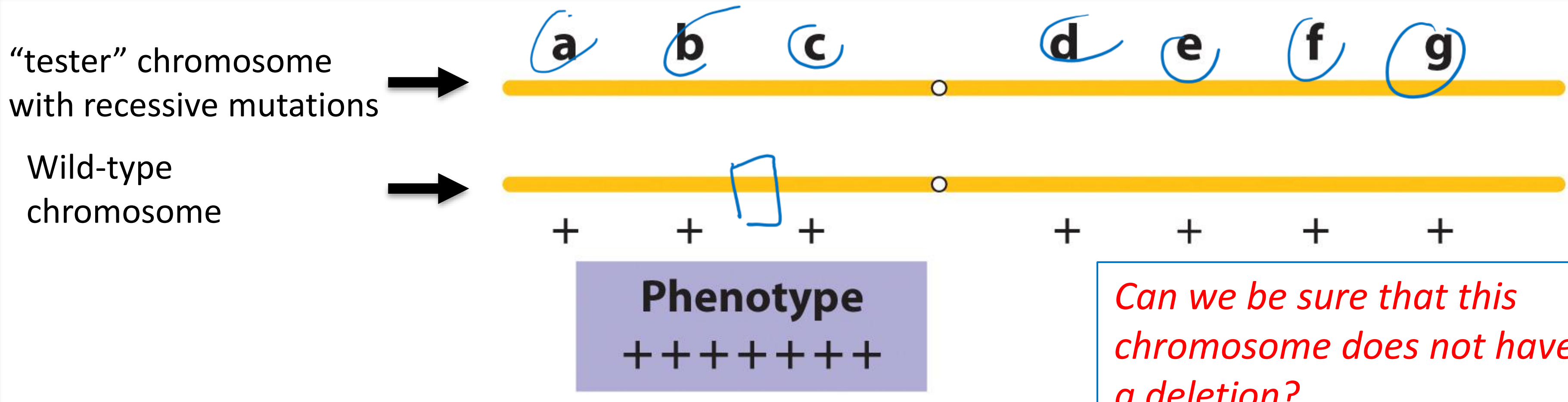
"tester" strain with multiple with recessive mutations

++++++/++++++ x abcdefg/abcdefg

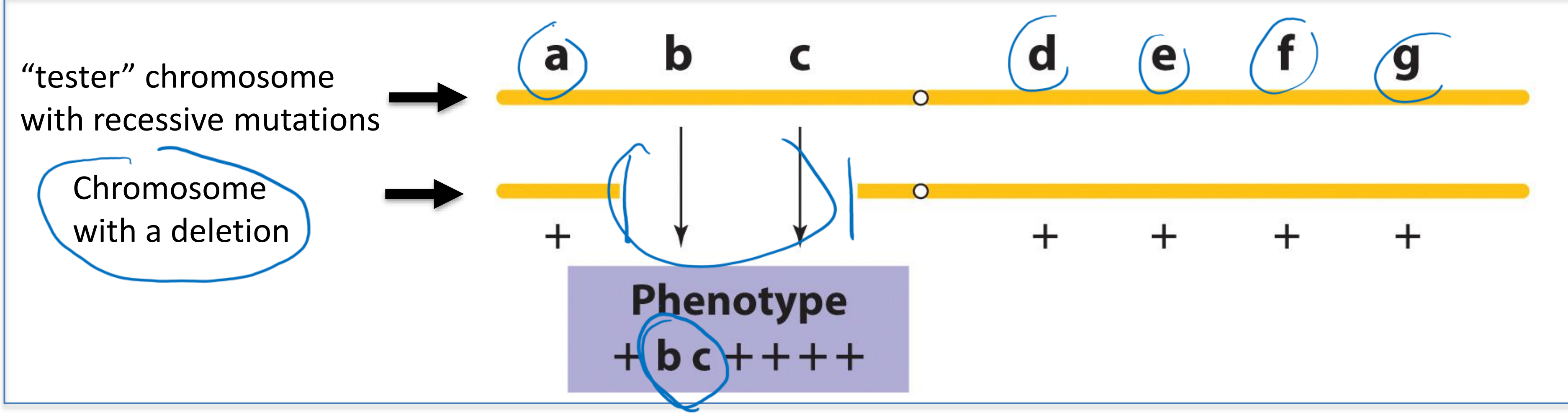
++++++/abcdefg Phenotype?

Detecting chromosomal deletions

complementation test



Can we be sure that this chromosome does not have a deletion?



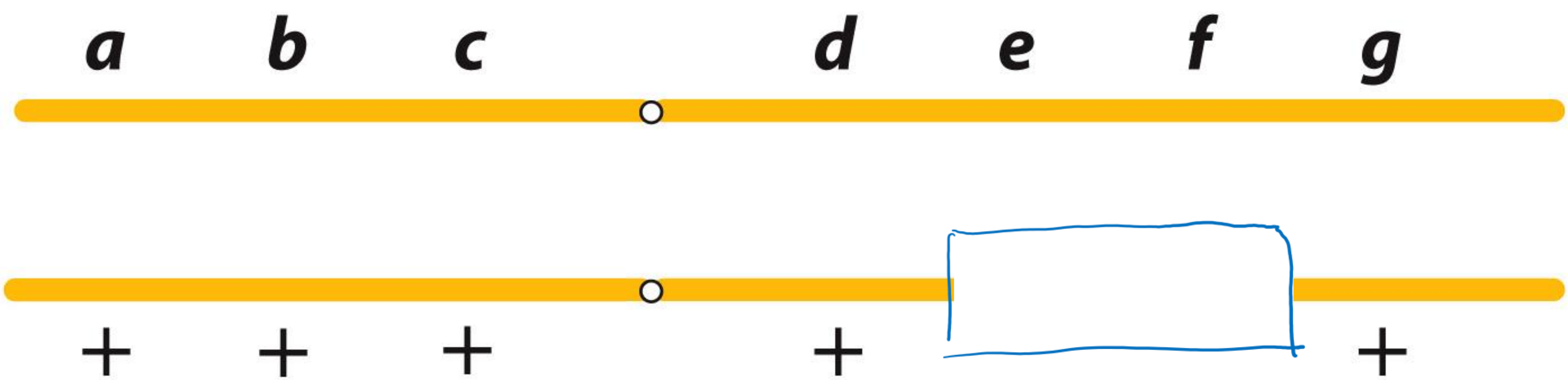
Using a complementation test to detect chromosomal deletions

"unknown strain" x abcdefg/abcdefg

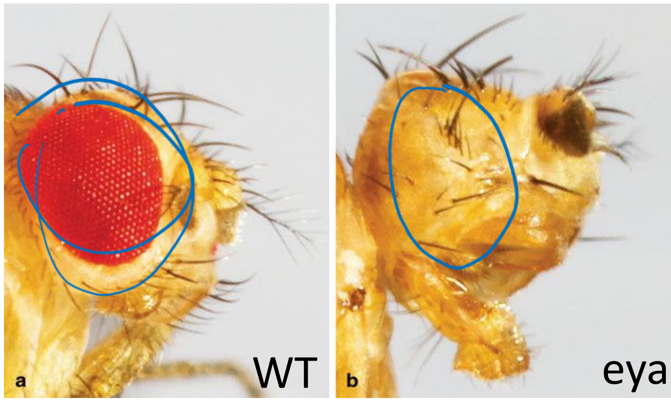
⬇

"unknown strain" /abcdefg

Phenotype: e f



Known deletions can be used to map a recessive mutant allele

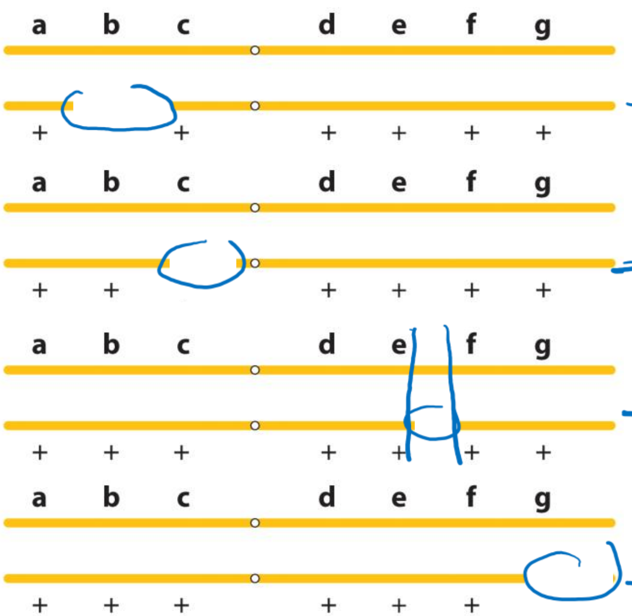


Example:

Flies homozygous for the “eya” (eyes absent) chromosome have no eyes.

Where is the mutated eya gene located?

Deletion mapping of the *eya* mutation: cross to known deletions and check F1 phenotype



phenotype: normal eyes

phenotype: normal eyes

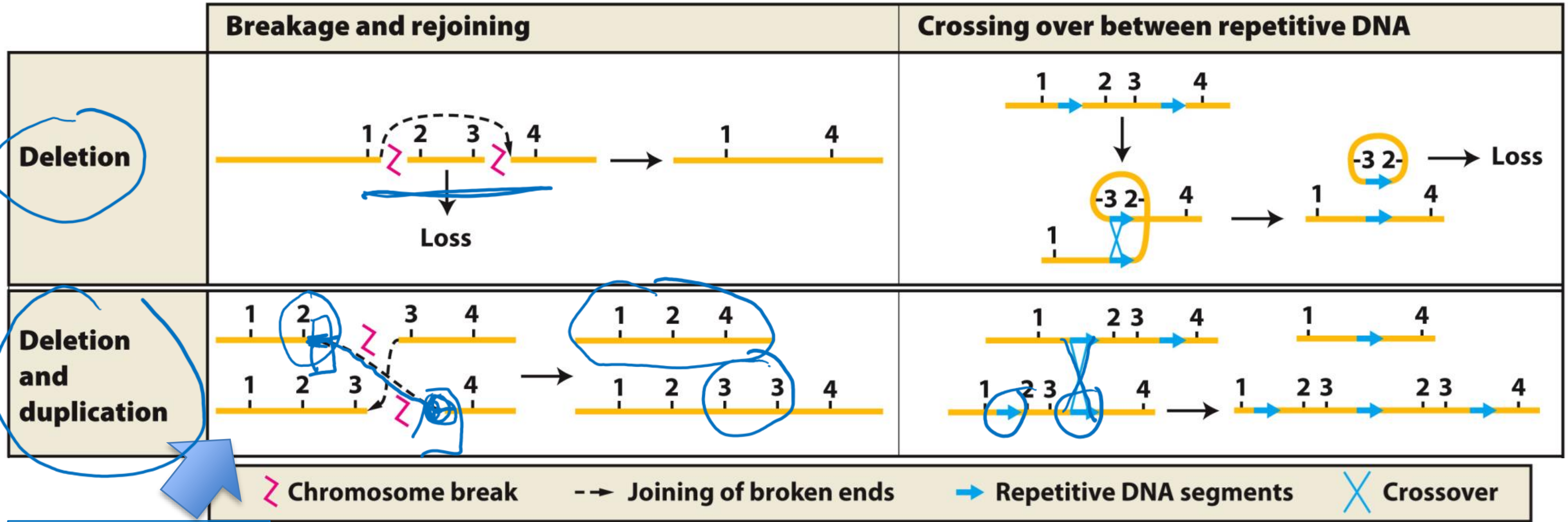
phenotype: no eyes

phenotype: normal eyes

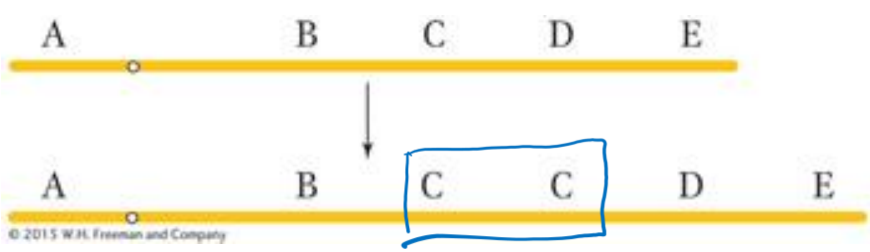
This analysis identifies which region of the chromosome bears the *eya* mutation.

Chromosomal Rearrangements:

Duplications



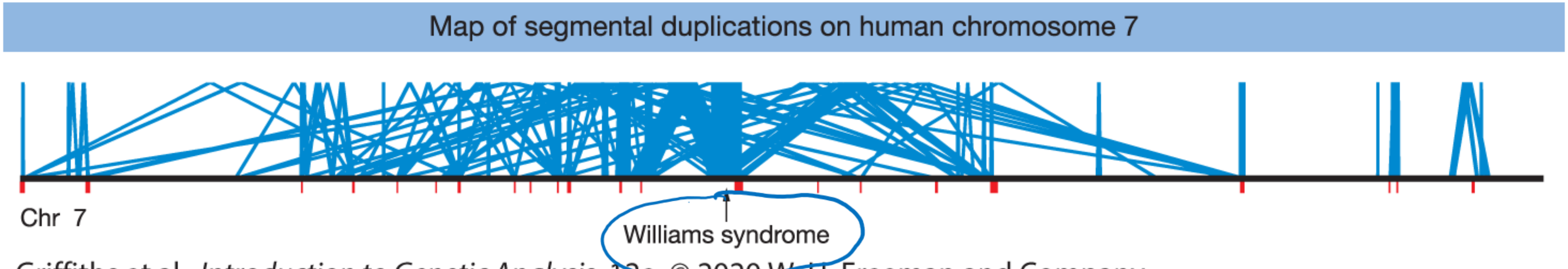
breaks on homologous chromosomes



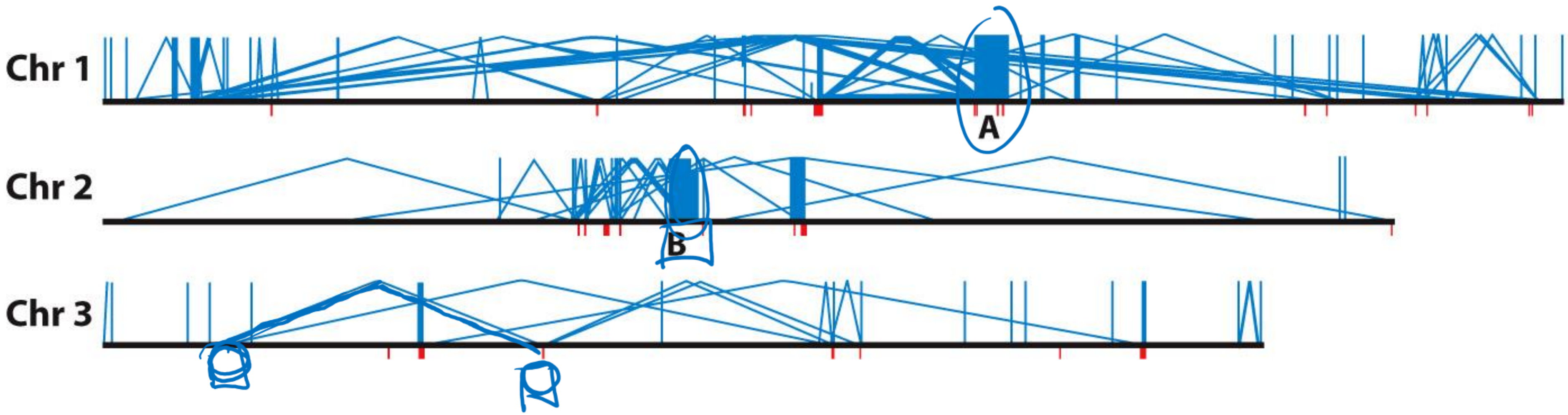
large duplications are called "segmental duplications"

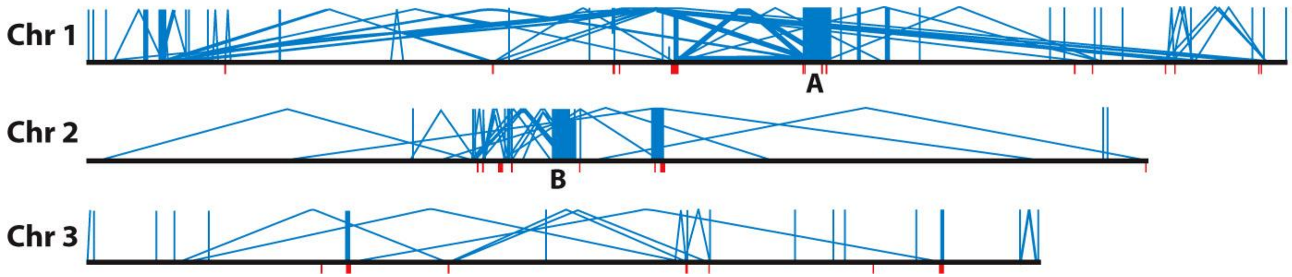
Chromosomal duplications

Map of segmental duplications in the human genome



Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company



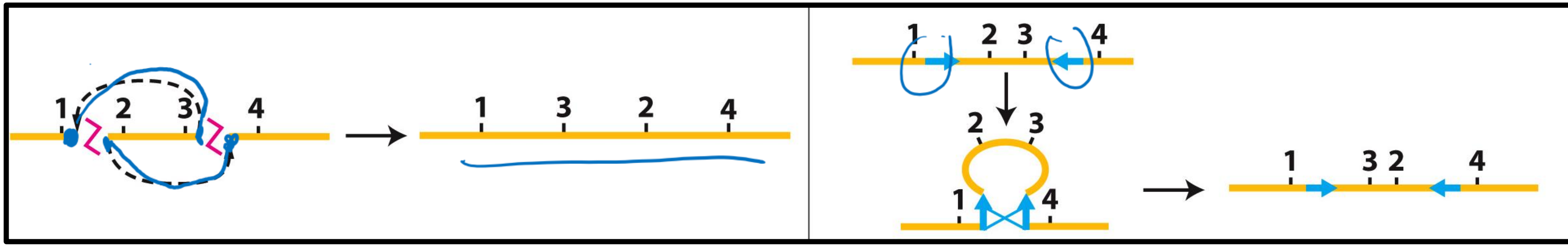


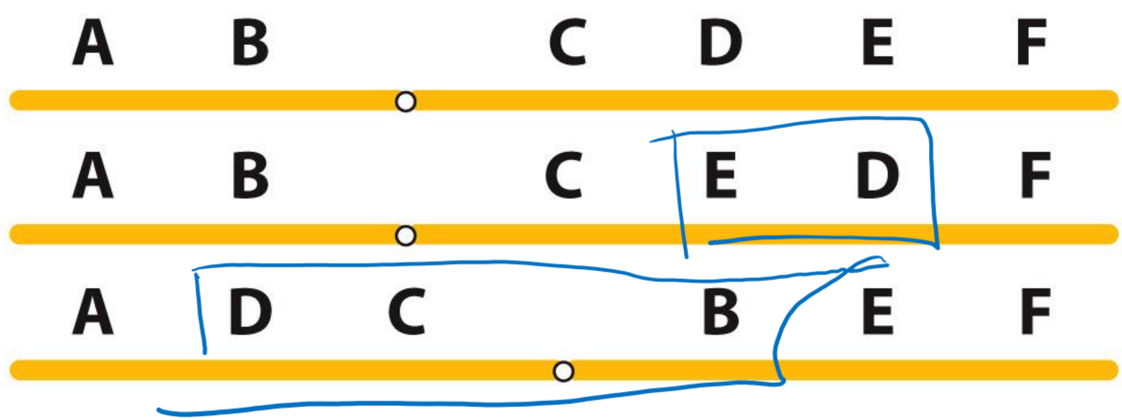
Think Break

Considering the profile of duplications in the genome and the mechanism for generating them, what can you conclude?

Chromosomal Rearrangements:

Inversions

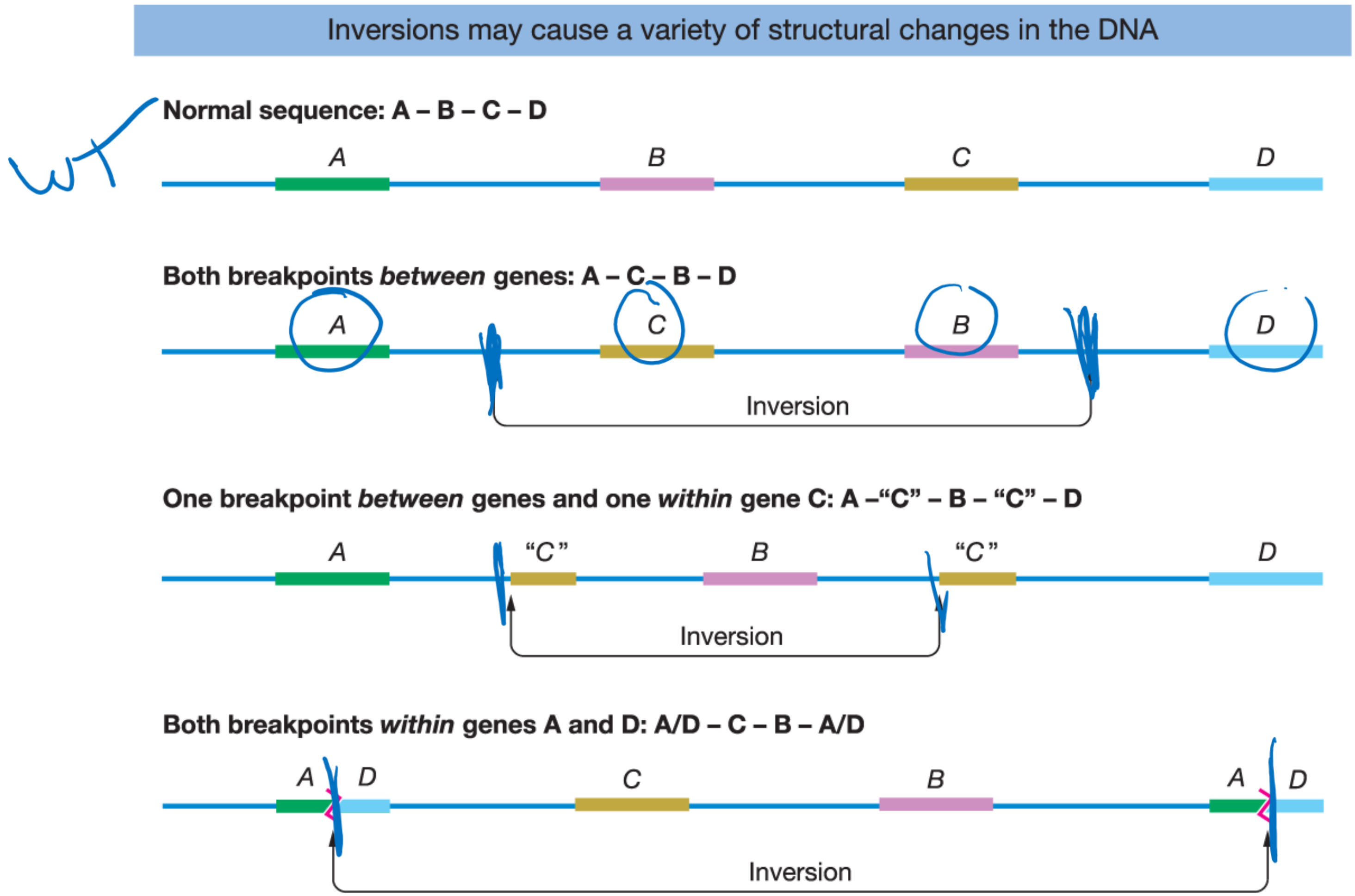




Unnumbered 17 p642
Introduction to Genetic Analysis, Eleventh Edition

Chromosomal inversions

Consequences: somatic



No genes disrupted

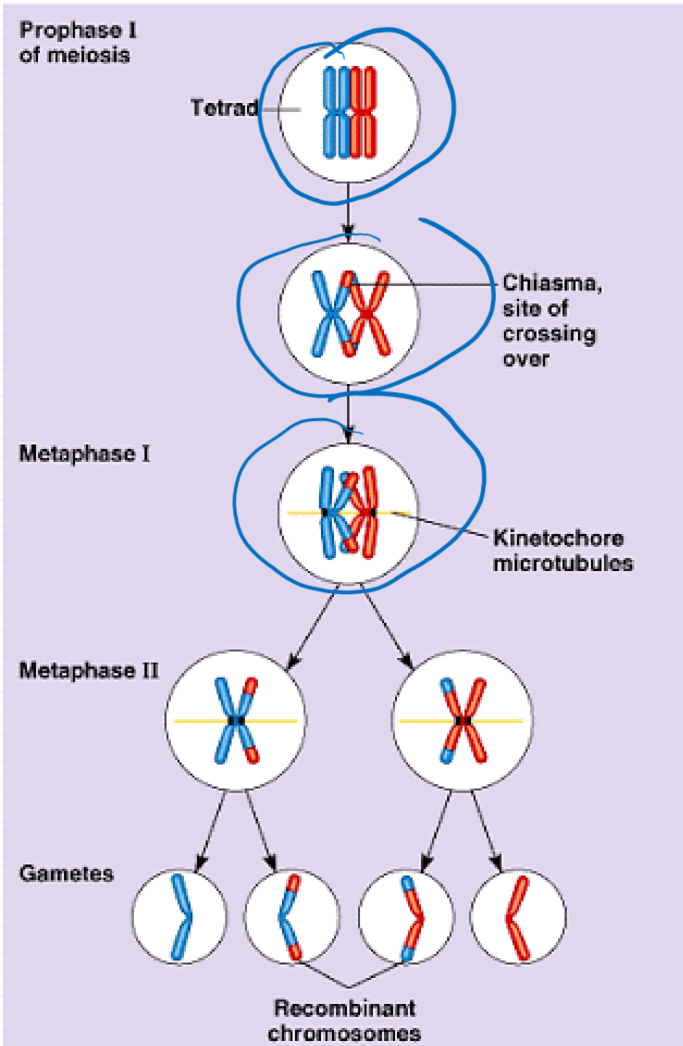
Gene C disrupted

A/D gene fusion

Griffiths et al., *Introduction to Genetic Analysis*, 12e, © 2020 W. H. Freeman and Company

Chromosomal inversions

Consequences: germline



Reminder: recombination occurs during meiosis I

Chromosomal inversions

Effects on chromosome behavior during meiosis

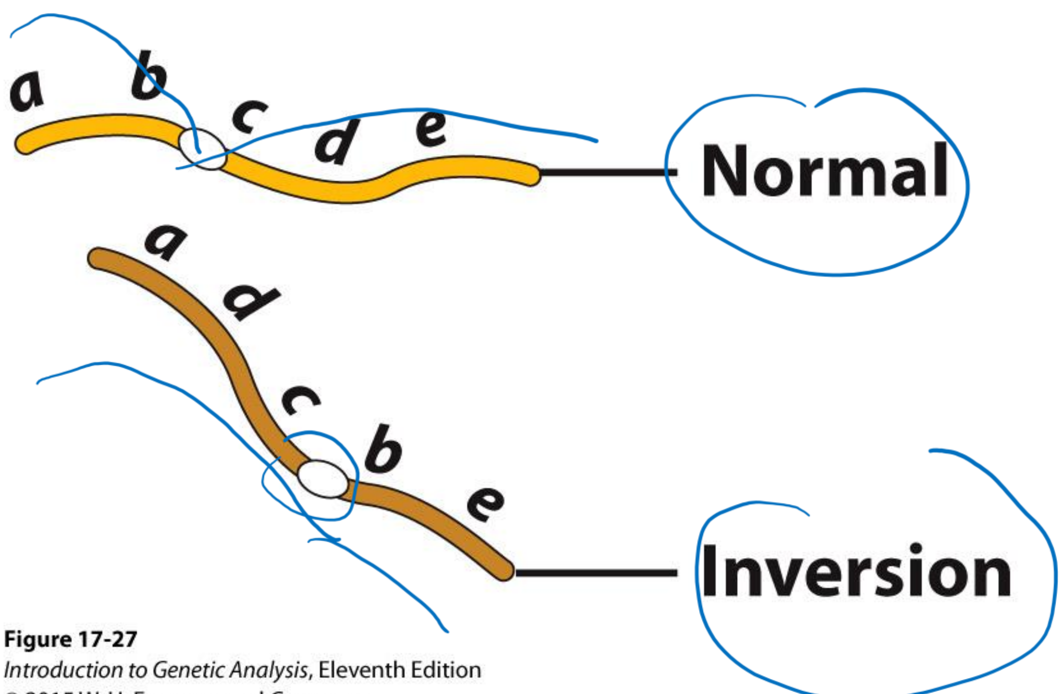
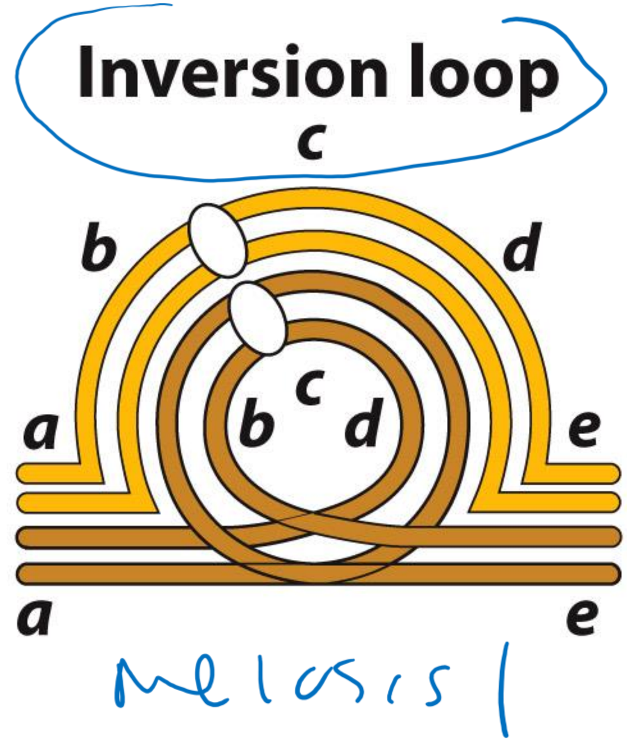
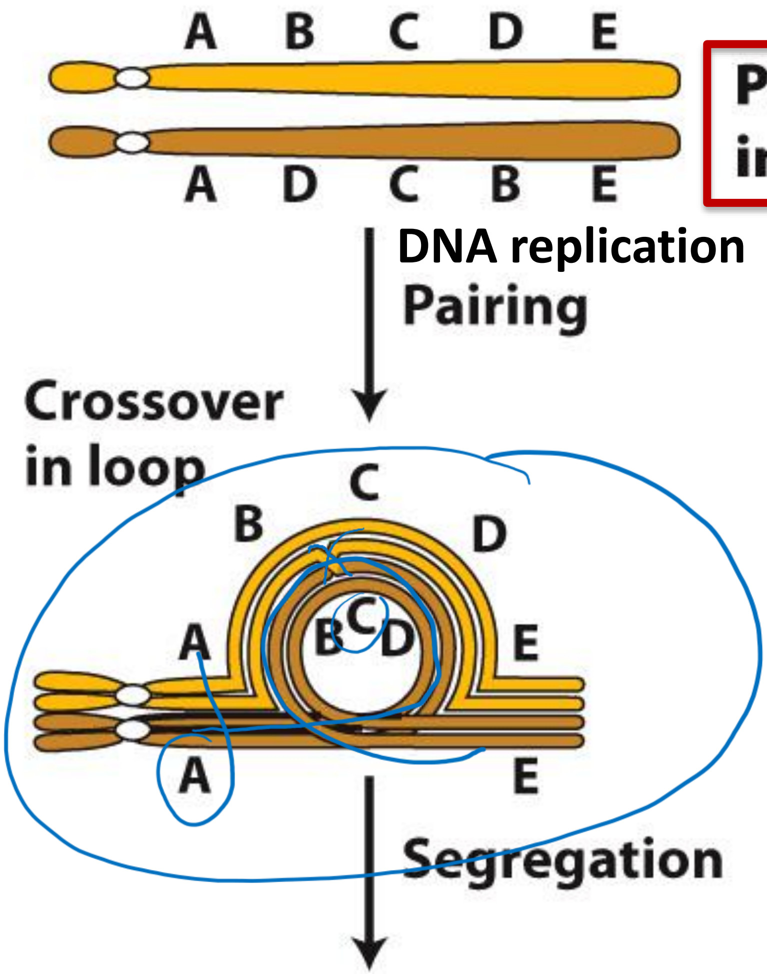


Figure 17-27
Introduction to Genetic Analysis, Eleventh Edition



Chromosomal inversions

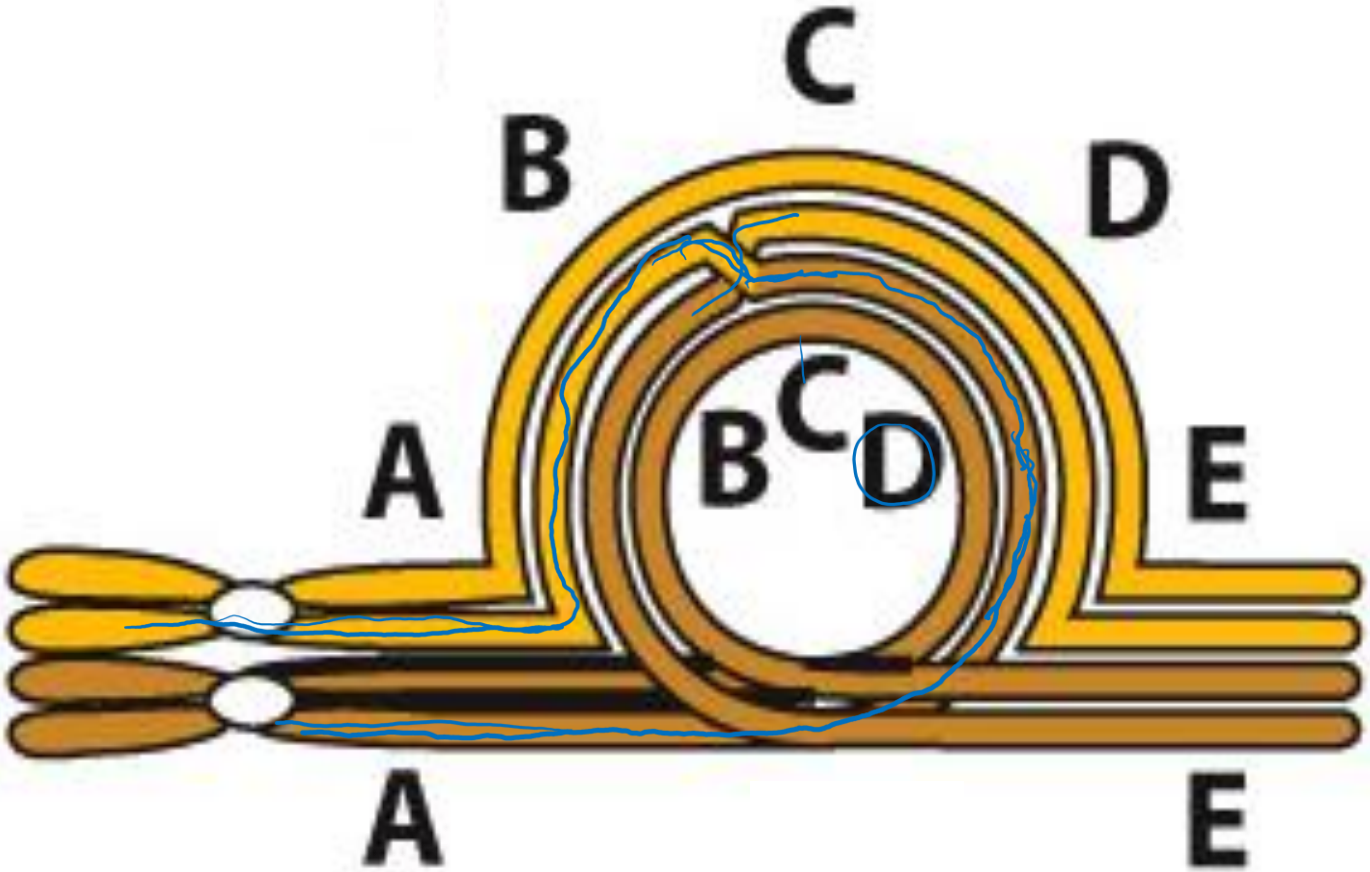
Effects on chromosome behavior during meiosis



Paracentric-inversion heterozygote

Chromosomal inversions

Consequences: germline



Crossing over in a paracentric inversion heterozygote → dicentric chromosome

Paracentric inversions → deletions

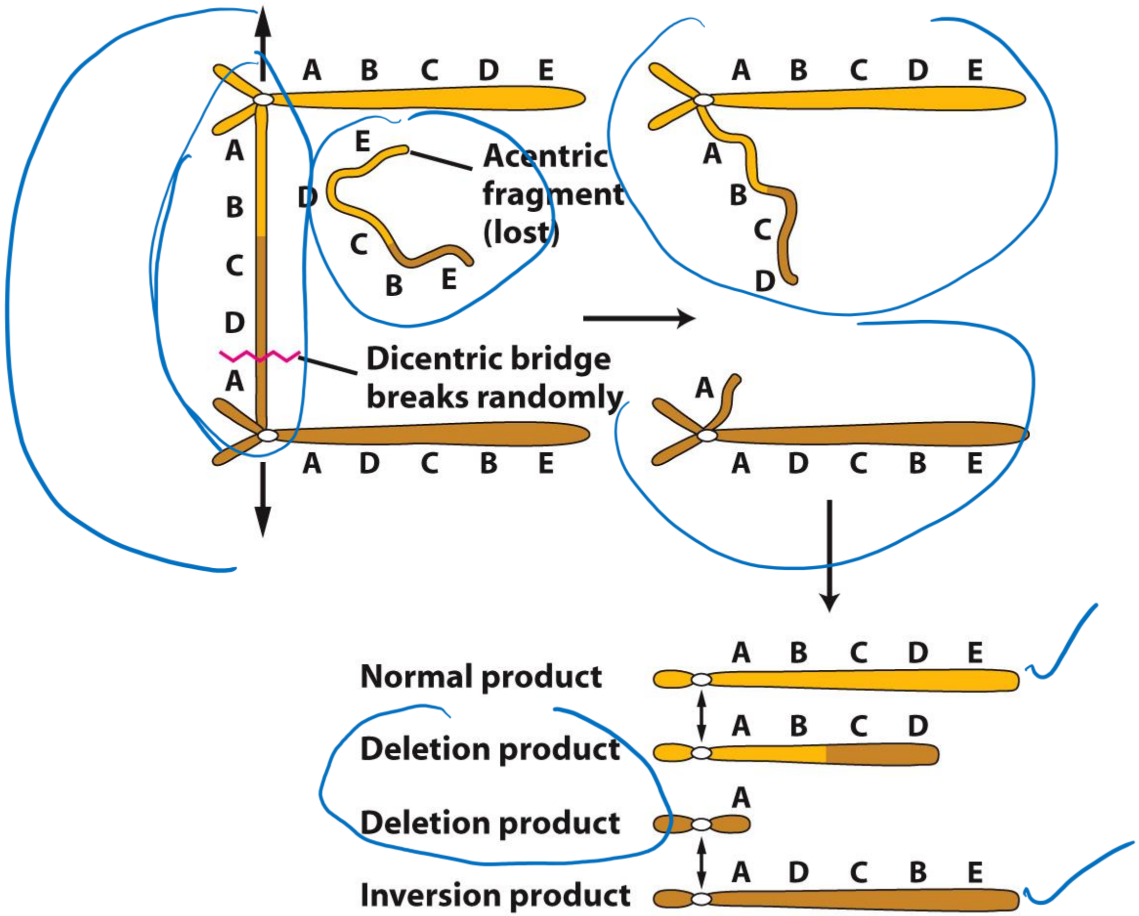


Figure 17-28 part 2
Introduction to Genetic Analysis, Eleventh Edition

Crossing over in a paracentric inversion heterozygote → dicentric chromosome → breakage → loss of acentric fragment and products with major deletions

Pericentric inversions → duplications and deletions

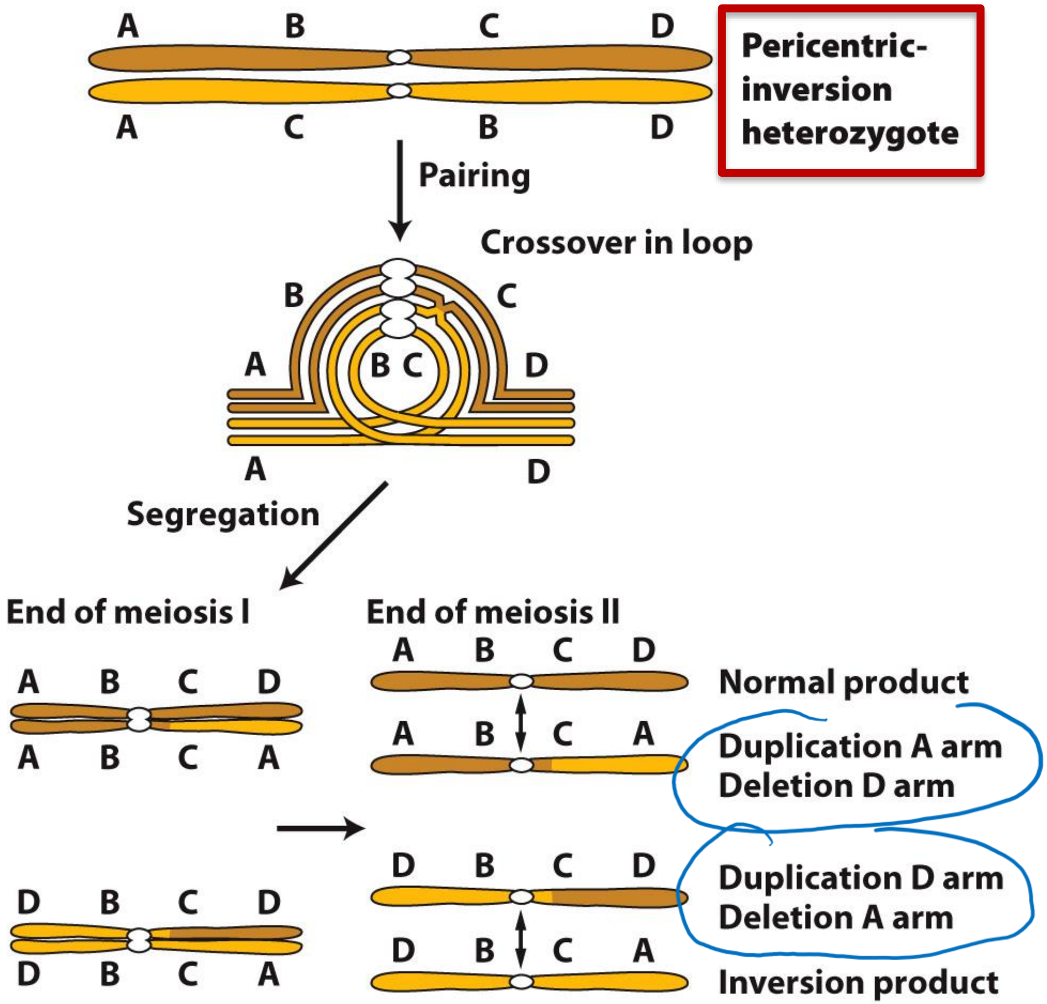
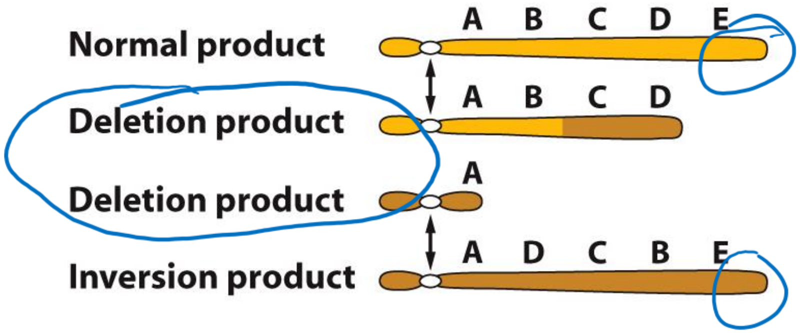
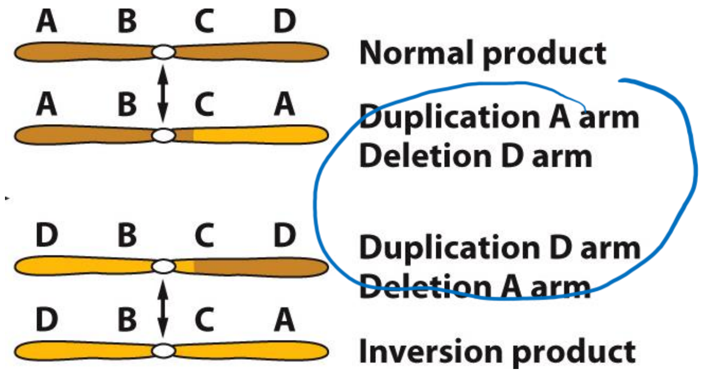


Figure 17-29
Introduction to Genetic Analysis, Eleventh Edition

Crossing over in a pericentric inversion heterozygote → products with major deletions meiotic products: paracentric inversion



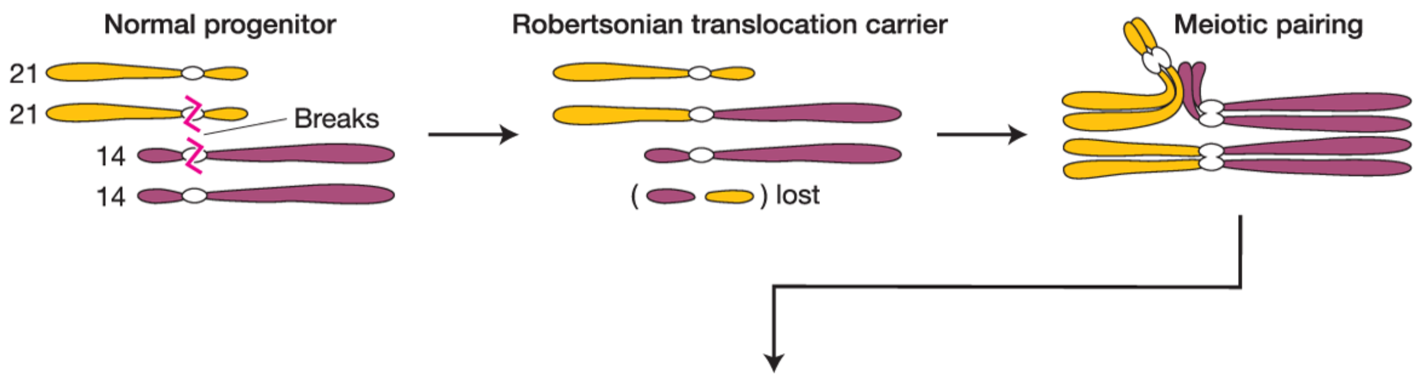
meiotic products: pericentric inversion



Think Break

What clinical consequences might you expect in an adult individual with a chromosomal inversion?

A balanced Robertsonian translocation resulting in inheritance of trisomy 21 (Down's Syndrome)



Segregation during meiosis → 6 possible gametes

A balanced Robertsonian translocation resulting in inheritance of trisomy 21 (Down's Syndrome)